Lecture 6: Trigonometric Integrals

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Integrating powers of sines and cosines. $\int \sin^m x \cos^n x \, dx$
- ☐ Integrating powers of $\tan x$ and $\sec x$. $\cot x$ and $\csc x$
- ☐ Integrating multiple angles of sine and cosine. $\int \sin(ax)\cos(ax)\,dx$

Introduction

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions. sin cos tan csc sec cot

The general idea is to use the identities to 'transform' the integrals into the format that are easier to work with.

① Basic formula

② Substitution

③ Integration by part

④ Use identities

⑤ multiplying by a form of 1

e.g. $\int \sec x \, dx = \int \sec x \, \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$

⑥ Products of power of sine and cosine

⑦ Products of powers of $\tan x$, $\sec x$ ($\csc x$, $\cot x$)

⑧ Product of sine and cosine of multiple angles

⑨ Eliminating square roots

Integrating powers of sine and cosine

Integrals of the form $\int \sin^m x \cos^n x \, dx$ where m and n are nonnegative integers.

There are essentially two cases:
(a) either m or n (or both) is odd; (at least one of them is odd)
(b) both m and n are even.

* whichever is odd you break it into $\sin x (\sin^{m-1} x)$ or $(\cos x)(\cos^{n-1} x)$. Then, use identity + substitution.

Case 1: Either m or n (or both) is odd

- When m is odd and n is even

Write $\sin^m x = \sin x (\sin^{m-1} x)$. Since $m-1$ is even, $\sin^{m-1} x = (\sin^2 x)^{\frac{m-1}{2}}$

Use identity $\sin^2 x = 1 - \cos^2 x$ and substitute $u = \cos x$, $du = -\sin x \, dx$

- When m is even and n is odd

Write $\cos^n x = \cos x (\cos^{n-1} x)$. Since $n-1$ is even, $\cos^{n-1} x = (\cos^2 x)^{\frac{n-1}{2}}$

Use identity $\cos^2 x = 1 - \sin^2 x$ and substitute $u = \sin x$, $du = \cos x \, dx$

- When both m and n are odd

Either of the above method.

Integrating powers of sine and cosine

$$\int \sin^3 x \cos^2 x \, dx = \int \sin x (\sin^2 x) \cos^2 x \, dx$$

$$= -\int \sin x (1 - \cos^2 x) \cos^2 x \, dx$$

Let $u = \cos x$

$du = -\sin x \, dx$

$$= -\int (1 - u^2) u^2 \, du = -\int u^2 - u^4 \, du$$

$$= \int u^4 - u^2 \, du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

$\sin^0 x \cos^5 x$ (5: odd)

$$\int \cos^5 x \, dx = \int \cos x (\cos^4 x) \, dx$$

$$= \int \cos x (\cos^2 x)^2 \, dx$$

Let $u = \sin x$

$du = \cos x \, dx$

$$= \int \cos x (1 - \sin^2 x)^2 \, du$$

$$= \int (1 - u^2)^2 \, du = \int 1 - 2u^2 + u^4 \, du = u - \frac{2u^3}{3} + \frac{u^5}{5} + C$$

$$= \sin x - \frac{2\sin^3 x}{3} + \frac{\sin^5 x}{5} + C$$

Integrating powers of sine and cosine

Let $\int \sin^{3} x \cos^{5} x \, dx = I$ (3 odd, 5 odd)

(M1) $I = \int \sin x \sin^2 x \cos^5 x \, dx$

$= \int \sin x (1-\cos^2 x) \cos^5 x \, dx$

$= -\int (1-u^2) u^5 \, du$

$= \int u^7 - u^5 \, du$

$= \frac{u^8}{8} - \frac{u^6}{6} + C$

$= \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + C$

Let $u = \cos x$

$du = -\sin x \, dx$

(M2) $I = \int \sin^3 x \cos x \cos^4 x \, dx$

Let $u = \sin x$
$du = \cos x \, dx$

$= \int \sin^3 x \cos x (\cos^2 x)^2 \, dx$

$= \int \sin^3 x \cos x (1 - \sin^2 x)^2 \, dx$

$= \int u^3 (1 - u^2)^2 \, du$

$= \int u^3 (1 - 2u^2 + u^4) \, du$

$= \int u^3 - 2u^5 + u^7 \, du = \frac{u^4}{4} - \frac{2u^6}{6} + \frac{u^8}{8} + C$

$= \frac{\sin^4 x}{4} - \frac{\sin^6 x}{3} + \frac{\sin^8 x}{8} + C$

You can check that Exercise.

$$\frac{\sin^4 x}{4} - \frac{\sin^6 x}{3} + \frac{\sin^8 x}{8} = \frac{\cos^8 x}{8} + \frac{\cos^6 x}{6} + D.$$

$$\frac{\sin^4 x}{4} - \frac{\sin^6 x}{3} + \frac{\sin^8 x}{8} = \frac{(1-\cos^2 x)^2}{4} - \frac{(1-\cos^2 x)^3}{3} + \frac{(1-\cos^2 x)^4}{8}$$

$$\vdots$$

Case 2: Both m and n are even

$$\sin^2 x = \frac{1}{2}[1 - \cos 2x]$$

$$\cos^2 x = \frac{1}{2}[1 + \cos 2x]$$

Recall that

$$\cos 2x = 2\cos^2 x - 1$$
$$= 1 - 2\sin^2 x$$

$$\int \sin^{2}(\text{even}) x \cos^{4}(\text{even}) x \, dx = \int \sin^2 x (\cos^2 x)^2 \, dx$$

$$= \int \left[\frac{1}{2}(1 - \cos 2x)\right]\left[\frac{1}{2}(1 + \cos 2x)\right]^2 dx$$

($\frac{1}{2}(1-\cos 2x) = \sin^2 x$, $\frac{1}{2}(1+\cos 2x) = \cos^2 x$)

$$= \frac{1}{8}\int (1 - \cos 2x)(1 + 2\cos 2x + \cos^2 2x)\, dx$$

$$= \frac{1}{8}\int (1 + 2\cos 2x + \cos^2 2x - \cos 2x - 2\cos^2 2x - \cos^3 2x)\, dx$$

$$= \frac{1}{8}\int \left(1 + \cos 2x - \cos^2 2x - \cos^3 2x\right) dx$$

1 → x

$\cos 2x$ → $\frac{\sin 2x}{2}$

$\cos^2 2x$ → even power (case ii)

$\cos^3 2x$ → odd power (case i)

$$\int \cos^2 2x\, dx = \int \frac{1}{2}\left[1 + \cos 4x\right] dx$$

$$= \frac{1}{2}\int 1 + \cos 4x\, dx$$

$$= \frac{1}{2}\left[x + \frac{\sin 4x}{4}\right] + C_1$$

$$\int \cos^3 2x\, dx = \int \cos 2x \cos^2 2x\, dx = \frac{1}{2}\int \cos 2x \left(1 - \sin^2 2x\right) dx$$

Let $u = \sin 2x$

$du = 2\cos 2x\, dx$

$$= \frac{1}{2}\int (1 - u^2)\, du$$

$$= \frac{1}{2}\left[u - \frac{u^3}{3}\right] + C_2$$

$$= \frac{1}{2}\left[\sin 2x - \frac{\sin^3(2x)}{3}\right] + C_2$$

So, $\int \sin^2 x \cos^4 x \, dx = \frac{1}{8}\left[\frac{2x}{2} + \frac{\sin 2x}{2} - \frac{1}{2}\left(x + \frac{\sin 4x}{4}\right)\right.$

$\left. - \frac{1}{2}\left[\sin 2x - \frac{\sin^3 2x}{3}\right]\right] + C.$

$= \frac{1}{16}\left[2x + \sin 2x - x - \frac{\sin 4x}{4} - \sin 2x + \frac{\sin^3 2x}{3}\right]$

$+ C$

$= \frac{1}{16}\left[x - \frac{\sin 4x}{4} + \frac{\sin^3 2x}{3}\right] + C.$

Eliminating Square Roots

$$\int_0^{\pi/4} \sqrt{1+\cos 4x}\, dx$$

Use identity $\cos 2x = 2\cos^2 x - 1$

$$= \int_0^{\pi/4} \sqrt{1+(2\cos^2 2x - 1)}\, dx$$

$$= \int_0^{\pi/4} \sqrt{2\cos^2 2x}\, dx$$

$$= \sqrt{2} \int_0^{\pi/4} \sqrt{\cos^2 2x}\, dx$$

$0 \le x \le \frac{\pi}{4} \Rightarrow 0 \le 2x \le \frac{\pi}{2} \Rightarrow 0 \le \cos 2x \le 1$ ($\cos 2x \ge 0$)

$$= \sqrt{2} \int_0^{\pi/4} |\cos 2x|\, dx$$

$$= \sqrt{2} \int_0^{\pi/4} \cos 2x\, dx = \sqrt{2}\left[\frac{\sin 2x}{2}\right]_0^{\pi/4} = \frac{\sqrt{2}}{2}(1-0) = \frac{\sqrt{2}}{2}.$$

Integrating powers of $\tan x$ and $\sec x$

Use the identity

$$\tan^2 x + 1 = \sec^2 x$$

to integrate by substitution or by parts when necessary.

Recall:

$\frac{d}{dx}\tan x = \sec^2 x$

$\frac{d}{dx}\sec x = \tan x \sec x$

$$\int \tan^4 x\,dx = \int (\tan^2 x)(\tan^2 x)\,dx$$

$$= \int (\tan^2 x)(\sec^2 x - 1)\,dx$$

$$= \int \underbrace{\tan^2 x \sec^2 x\,dx}_{\text{substitute } u=\tan x,\ du=\sec^2 x\,dx} - \int \underbrace{\tan^2 x\,dx}_{\sec^2 x - 1}$$

$$= \int u^2\,du - \int \sec^2 x - 1\,dx = \frac{u^3}{3} - (\tan x - x) + C$$

$$= \frac{\tan^3 x}{3} - \tan x + x + C$$

Integrating powers of $\tan x$ and $\sec x$

$$\int \sec^3 x\,dx = \int \sec x \sec^2 x\,dx$$

Let $u = \sec x$, $dv = \sec^2 x\,dx$

$du = \tan x \sec x\,dx$, $v = \tan x$

(easy to integrate: $\frac{d}{dx}\tan x = \sec^2 x$)

by parts

$$= \sec x \tan x - \int \tan x \tan x \sec x\,dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1)\sec x\,dx$$

$$\int \tan^4 x \sec^4 x\,dx = \sec x \tan x - \int \sec^3 x\,dx + \int \sec x\,dx$$

$$2\int \sec^3 x\,dx = \sec x \tan x + \int \sec x\,dx$$

$$\int \sec^3 x\,dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\int \sec x\,dx$$

$$= \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

Exercise: By multiplying $\frac{\sec x + \tan x}{\sec x + \tan x}$, show that $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

$$\int \tan^4 x \sec^4 x \, dx = \int \tan^4 x \, (\sec^2 x)(\sec^2 x) \, dx$$

$$= \int \tan^4 x \, (1 + \tan^2 x)(\sec^2 x) \, dx$$

Let $u = \tan x$

$du = \sec^2 x \, dx$

$$= \int u^4 (1 + u^2) \, du$$

$$= \int u^4 + u^6 \, du$$

$$= \frac{u^5}{5} + \frac{u^7}{7} + C$$

$$= \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + C.$$

Integrating multiple angles of sine and cosine

Integrals of the form

$$\int \sin mx \sin nx \, dx, \quad \int \sin mx \cos nx \, dx, \quad \text{and} \quad \int \cos mx \cos nx \, dx.$$

Use the identities

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x],$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m-n)x + \sin(m+n)x],$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m-n)x + \cos(m+n)x].$$

Recall: $\sin(a+b) = \sin a \cos b + \cos a \sin b$ — ①

$\sin(a-b) = \sin a \cos b - \cos a \sin b$ — ②

$\cos(a+b) = \cos a \cos b - \sin a \sin b$ — ③

$\cos(a-b) = \cos a \cos b + \sin a \sin b$ — ④

To get ⊛, we take ① + ②

$$\sin(a+b) + \sin(a-b) = 2\sin a \cos b$$

$$\sin a \cos b = \frac{1}{2}\left[\sin(a+b) + \sin(a-b)\right]$$

Let $a = mx$ and $b = nx$,

$$\sin(mx)\cos(nx) = \frac{1}{2}\left[\sin(mx+nx) + \sin(mx-nx)\right]$$

$$= \frac{1}{2}\left[\sin(m-n)x + \sin(m+n)x\right]$$

Exercise: Try to get the other two identities from ①, ②, ③, ④.

Integrating multiple angles of sine and cosine

$$\int \sin 3x \cos 5x \, dx = \int \frac{1}{2}\left[\sin(3x+5x) + \sin(3x-5x)\right] dx$$

$$= \frac{1}{2}\int \sin 8x + \sin(-2x)\, dx$$

$$= \frac{1}{2}\int \sin 8x - \sin 2x \, dx, \text{ since sine is an odd function.}$$

$$= \frac{1}{2}\left[\frac{-\cos 8x}{8} - \frac{-\cos 2x}{2}\right] + C = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C.$$

$$\int \cos 5x \cos 3x \, dx = \int \frac{1}{2}\left[\cos(5x+3x) + \cos(5x-3x)\right] dx$$

$$= \frac{1}{2}\int \cos 8x + \cos 2x \, dx$$

$$= \frac{1}{2}\left[\frac{\sin 8x}{8} + \frac{\sin 2x}{2}\right] + C$$

$$= \frac{\sin 8x}{16} + \frac{\sin 2x}{4} + C.$$

Integrating multiple angles of sine and cosine

$$\int \sin 3x \sin 2x \, dx = \int \frac{1}{2}\left[\cos(3x-2x) - \cos(3x+2x)\right] dx$$

$$= \frac{1}{2}\int \cos x - \cos 5x \, dx$$

$$= \frac{1}{2}\left[\sin x - \frac{\sin 5x}{5}\right] + C$$

$$= \frac{\sin x}{2} - \frac{\sin 5x}{10} + C$$